CNSQ DESKTOP APPLICATION SPRINT

7-Working-Day Implementation & Delivery Roadmap (Tauri v2)

CONNECT NIGERIA

Timeline: 7 Working Days

EXECUTIVE SUMMARY

- Objective: Package existing CNSQ Web App into a native desktop application (Windows & macOS).
- Architecture: Zero rewrite needed - preserves 100% of Next.js frontend and NestJS backend APIs.
- Technology: Tauri v2 - ultra-lightweight installers (~15 MB) and high efficiency (~40 MB RAM).

7-WORKING-DAY SPRINT SCHEDULE & DELIVERABLES

TIMELINE	FOCUS AREA	KEY DELIVERABLES & OBJECTIVES
Day 1	**Environment & POC**	- Initialize Tauri v2 framework inside the codebase. - Compile and launch local native desktop window running CNSQ. - Milestone: Verified local desktop window rendering CNSQ.
Day 2	**Routing & Auth Adaptation**	- Validate authentication flows (JWT tokens, sessions, persistent storage). - Configure routing and smooth API communication with NestJS backend. - Ensure external URLs open cleanly in user's default web browser.
Day 3	**Desktop Window & Branding**	- Embed official CNSQ icons (Windows .ico and macOS .icns). - Implement native window handling & minimize-to-system-tray behavior. - Configure auto-launch on system startup capability.
Day 4	**Native OS Notifications**	- Implement native OS desktop push alerts for tickets and chat messages. - Configure system tray notification badge / unread counters. - Verify background alert delivery when app window is minimized.
Day 5	**Optimization & Polish**	- Conduct memory and performance profiling (target: under 80 MB RAM). - Optimize startup speed and window launch state transitions. - UI polish for desktop scaling across high-DPI and standard displays.
Day 6	**Build & Cross-Platform QA**	- Generate standalone installers (.exe/.msi for Windows, .dmg for macOS). - Multi-platform test on Windows and macOS test machines. - Conduct internal QA validation covering ticketing and chat workflows.
Day 7	**Testing Review & Live Demo**	- Final bug fixes and edge-case testing based on QA feedback. - Prepare documentation and test installation files for stakeholders. - Milestone: Deliver production-ready installers & present live demo.

CORE BUSINESS VALUE

Independent Native Software:
Real installable app (.exe / .dmg), not a PWA.

Zero Interruption to Web Users:
Web and Desktop share the same codebase.

System Presence:
Icon on desktop, dock, system tray, & startup.

Speed & Efficiency:
Under 15 MB installer size, under 80 MB RAM.

POST-SPRINT ROADMAP (PHASE 2)

Code Signing & Trust:
Apple Developer & Windows Authenticode certs to eliminate 'Unknown Publisher' security warnings.

Background Auto-Updater:
Seamlessly pushes future releases to employees.

CI/CD Build Automation:
GitHub Actions workflow to compile binaries.

Granular Offline Caching:
Local caching for offline ticket drafting.